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Modeling Nonlinear Insulin-Glucose Dynamics in Critically-ill Patients

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Abstract

Tight glycaemic control for critically ill patients in the intensive care unit is crucial to prevent adverse outcomes influenced by metabolic fluctuations associated with insulin-resistance. Traditional insulin intensive therapy in patients has been demonstrated to inadequately sustain glucose concentration in a safe and stable range. Utilizing clinically-validated models allows for modeling unknown insulin-glucose dynamics with state estimation methods. Previous work in estimating insulin-glucose dynamics failed to consider a diverse synthetic patient cohort with differing nutrition inputs, as well as motivation for the parameterization of their proposed filters. In this project, a framework to sample synthetic patients was developed to model critically ill patients residing in the Karolinska University Hospital, enabling estimation of insulin-glucose dynamics. The Intensive Control Insulin-Nutrition-Glucose model used in this study and previous work was found to not be observable when computing the empirical observability gramians, further highlighting the limitations of modeling glucoregulation from blood glucose measurements alone. The extended Kalman filter was identified to struggle in estimating blood glucose with large deviations from the true state in the initial prediction ($\pm 10\%$ and $\pm 25\%$) in comparison to the particle filter, but was measured to be run significantly faster. A particle filter with 750 particles achieved the best accuracy-complexity tradeoff, yielding more than a 6 times reduction in complexity from the previous method using 5000 particles. To extend upon previous work, estimation of a parameter not represented in the Intensive Control Insulin-Nutrition-Glucose model formulation was performed. The particle filter managed to estimate both constant and time-varying insulin sensitivity, while the extended Kalman filter only reliably estimated the former. Consequently, drawing attention to the limitations of the linearization approximations seen in the extended Kalman filter. The methodology proposed in this study allows for clinicians to follow a systematic process to develop their own model-based tight glycaemic control method.

Keywords

Bayesian filter, Blood glucose, Critically ill patients, Insulin-resistance, Insulin-glucose dynamics, Insulin sensitivity, Model-based control, Tight glycemic control

Sammanfattning

Tight glycemic control för kritiskt sjuka patienter på intensivvårdsavdelning är avgörande för att förebygga ogynnsamma utfall som påverkas av metabola fluktuationer i samband med insulinresistens. Traditionell intensiv insulinbehandling har visats inte tillräckligt kunna upprätthålla glukoskoncentrationen inom ett säkert och stabilt intervall. Genom att använda kliniskt validerade modeller möjliggörs skattning av okända insulin–glukos dynamiker med hjälp av tillståndsskattningsmetoder. Tidigare arbete med skattning av insulin–glukos-dynamik tog inte hänsyn till en divers syntetisk patientkohort med olika nutritionsinsatser, och motiveringen för parameteriseringen av det föreslagna filtret var bristfälligt beskriven. I detta projekt utvecklades ett ramverk för att sampla syntetiska patienter som modellerar kritiskt sjuka patienter vid Karolinska Universitetssjukhuset, vilket möjliggjorde skattning av insulin–glukos dynamik. Intensive Control Insulin-Nutrition-Glucose modellen som användes i denna studie och i tidigare arbete visade sig inte vara observerbar vid beräkning av de empiriska observerbarhetsgramianerna, vilket ytterligare belyser begränsningarna med att modellera glukosreglering enbart utifrån blodglukosmätningar. Det identifierades att det utökade Kalmanfiltret hade svårigheter att skatta blodglukos när den initiala prediktionen avvek kraftigt från det sanna tillståndet ($\pm 10\%$ och $\pm 25\%$) jämfört med partikelfiltret, men att det ändå kunde köras betydligt snabbare. Ett partikelfilter med 750 partiklar uppnådde den bästa avvägningen mellan noggrannhet och komplexitet och gav mer än en sexfaldig minskning i komplexitet jämfört med den tidigare metoden med 5000 partiklar. För att vidareutveckla tidigare arbete skattades även en parameter som inte ingår i Intensive Control Insulin-Nutrition-Glucose modellens formulering. Partikelfiltret lyckades uppskatta både konstant och tidsvarierande insulinkänslighet, medan den utökade Kalmanfiltret endast tillförlitligt uppskattade den förra. Detta belyser begränsningarna hos de linjäriseringsapproximationer som används i det utökade Kalmanfiltret. Metodiken som introduceras i denna studie kan hjälpa kliniker att följa en systematisk process för att utveckla egna modellbaserade metoder för tight glycemic control.

Nyckelord

Bayesiskt filter, Blodglukos, Insulin–glukos dynamiker, Insulinkänslighet, Insulinresisten, Kritiskt sjuka patienter, Modellbaserade kontrollera, Tight glycemic control

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Acronyms

ICING	Intensive Control Insulin-Nutrition-Glucose
IPI	Integral-based Parameter Identification
MSE	Mean squared error

Chapter 1

Introduction

Patients in the intensive care unit suffering from critical illnesses (e.g. acute myocardial infarction, sepsis, stroke) have been widely documented to undergo hyperglycaemia, even those without a prior diagnosis of diabetes [1]. Acute stress observed in critical illnesses often induces physiological changes, as the secretion of stress hormones results in an inflammatory response associated with insulin-resistance [2] [3] [4] [5] [6] [7]. While there is inconclusive evidence pertaining to the causality of stress hyperglycaemia and adverse outcomes, the onset of mortality is far greater in patients with no reported history of hyperglycaemia [1] [3]. Sustaining glucose concentration under 6.106 mmol/L through insulin infusion has demonstrated a reduction of mortality and morbidity, compared to less-intensive methods of glycaemic control [8], at the subsequent cost of an increase of reported incidences of hypoglycaemia [9]. Further review of intensive insulin therapy and tight glucose control has identified discrepancies in the administration of glycaemic control, mostly corresponding to the risk of moderate to severe hypoglycaemia and glycaemic variability in a critically ill patient is equally detrimental as stress hyperglycaemia, with an increase of mortality and health complications observed [9] [10] [11].

1.1 Background

Clinicians now aim to maintain blood glucose in a range of 6 mmol/L to 10 mmol/L to account for adverse outcomes in hyperglycaemia and hypoglycaemia. Ad hoc sliding protocols have been utilized in the intensive care unit to provide clinicians with the necessary nutrition and insulin doses to control blood glucose levels based on hourly glucose measurements [12]

[13]. Such methods are often difficult to implement in practice and result in minimal improvements in preventing mortality [14]. The implementation of tight glucose controllers avoids tedious procedures followed in sliding protocols by estimating insulin-glucose states on given insulin and nutrition inputs, with observed improvements in constraining blood glucose to a safe range [15]. The implementation of model-based tight glucose control methods relies on the formulation of insulin-glucose dynamics. Mechanistic insulin-glucose models have been developed with an extensive physiological foundation, and have been validated for glycaemic control in clinical settings [14] [15] [16] [17] [18]. The insulin-glucose models derived are nonlinear, which requires more advanced numerical methods to properly estimate the current and future insulin-glucose states in patients. In practice, the estimated states from the insulin-glucose models are affected by process noise from unmodeled states (and/or parameters) in the model formulation and from noisy measurements. As a direct result, the insulin-glucose states in practice struggle to be estimated deterministically. To accommodate modeling of stochastic processes, statistical methods are needed to optimally estimate the states in mechanistic insulin-glucose models.

1.2 Problem

Patients suffering with differing critical illnesses are observed with varying magnitudes of insulin resistance, preventing a one-size-fits-all tight glucose control protocol to restrict glycaemic levels. Model-based tight glucose control has the ability to adapt to a given patient's condition, but there currently exist gaps in research for in-depth analysis of statistical modeling of insulin-glucose states. Previous studies have employed Bayesian filtering methods for this task [19] [20] and integral-based approximation to estimate unknown insulin-glucose model parameters [18]. These studies lack systematic design of the synthetic patient cohort and relationship of filter configuration and performance of tight glucose control. The gap in research in analyzing various Bayesian filters with representative synthetic patients hinders gauging how well tight glucose controllers work in practice with vulnerable patients.

1.3 Purpose

This thesis aims to improve the current research and implementation of tight blood glucose control. Researchers studying insulin-glucose systems in critically ill patients will benefit from a cohesive and reproducible methodology to accommodate their own research goals. Better glycaemic control will result in clinicians being able to adjust nutrition and insulin doses from the prediction of insulin-glucose states in between hourly measurements. Consequently, giving clinicians crucial time to better prevent mortality in patients susceptible to hypoglycaemia or hyperglycaemia.

1.4 Goals

The overall goals of this project has been divided into the following sub-goals:

1. Design a synthetic patient cohort
2. Compare performance of various filters in estimation of insulin-glucose states
3. Estimate unknown insulin-glucose model parameters in patients

1.5 Research Methods

The modeling of the insulin-glucose dynamics found in critically ill patients will be implemented on a synthetic patient cohort. Using synthetic patients allows the researcher to manually control the insulin-glucose model parameters, which directly improves the evaluation of the outcome of the insulin-glucose control. Previous work will serve as the benchmark to compare how the parameterization of our method performs in estimating insulin-glucose dynamics of synthetic patients. The chosen filter design will then be used to estimate additional dynamics not mathematically represented in mechanistic insulin-glucose models. Fitting metrics and computational complexity of the estimators will be utilized to draw inferences about the performance.

1.6 Delimitations

This study will consist of a synthetic patient cohort. Using synthetic patients is important for maintaining internal validity in the experimental process, as it verifies the observed outcome of the experiment is due to the manipulation of insulin-glucose model parameters rather than uncontrolled variables. The expense of using synthetic patients can be identified in the reduced generalizability and external validity of this experiment, since synthetic patients are reflections of model assumptions. While this is a limitation, this study will focus on the foundation of the modeling of insulin-glucose dynamics, and future work can replicate the study with real clinical data.

1.7 Structure of the thesis

Chapter 2 will introduce the existing modeling of insulin-glucose dynamics used, followed by an overview of discretization, state estimation, and related work in parameter estimation. In Chapter 3, the proposed research process used to answer the research questions of this study will be documented and motivated. The results and the subsequent discussion will be presented in Chapter 4. The key findings and future work of the experiment will be summarized in Chapter 5.

Chapter 2

Background

The background will introduce the theoretical concepts and prior work relevant to the thesis. Beginning with a description of the system physiology and the ICING model. Followed by presenting how state estimation methods can be applied to estimate insulin-glucose states optimally. The chapter will then conclude with practical parameter estimation approaches to predict unknown parameters.

2.1 Physiology of glucoregulation

To model glucoregulation in critically ill patients, it is necessary to understand how the physiological mechanisms are interrelated. Glucoregulation in the human body relies on the function of the pancreas. When there is a decrease in blood glucose, the alpha cells in the pancreas produce the hormone glucagon, that binds to liver cells. Resulting in the conversion of the stored glycogen in the liver to glucose, which is secreted in the bloodstream. In the case of an increase of blood glucose, such as the intake of a meal, beta cells in the pancreas will release insulin. The kidneys will filter the produced insulin, and the liver will be signaled to convert glucose into glycogen. Thus, decreasing the total amount of blood glucose.

As a result of the previously mentioned insulin resistance found in critically-ill patients, the body does not normally respond to insulin, resulting in lowered glycogen production in the liver. External food is given to patients to ensure proper nutrition, but food being digested in the body will result in a spike of blood glucose, as the glucose transfers from the stomach to the gut. As a result, it is imperative that patients with insulin resistance are given exogenous insulin by clinicians to maintain glycaemic ranges in a safe range.

Given these functions of the human glucoregulation system, a model can be developed to build intuition of the insulin-glucose dynamics in patients.

2.2 Intensive Control Insulin-Nutrition-Glucose Model

The ICING model presented by Lin et al. [17] captures the insulin-glucose dynamics in critically ill patients. The model is constructed as a fusion of two existing models validated for patients in the intensive care unit [14] [16], containing the following ensemble of differential equations

$$\dot{G} = -p_G G(t) - S_I(t)G(t)\frac{Q(t)}{1+\alpha_G Q(t)} + \frac{P(t)+E_G-C_G}{V_G} \tag{2.1a}$$

$$\dot{Q} = n_I(I(t)-Q(t)) - n_C\frac{Q(t)}{1+\alpha_G Q(t)} \tag{2.1b}$$

$$\dot{I} = -n_K I(t) - n_L\frac{I(t)}{1+\alpha_I I(t)} - n_I(I(t)-Q(t)) + \frac{u_{ex}(t)}{V_I} + \frac{(1-x_L)u_{en}(t)}{V_I} \tag{2.1c}$$

$$\dot{P_1} = -d_1 P_1 + D(t) \tag{2.1d}$$

$$\dot{P_2} = -\min\left(d_2 P_2, P_{max}\right) + d_1 P_1 \tag{2.1e}$$

$$u_{en}(t) = k_1 \exp\left(-\frac{k_2}{k_3}I(t)\right), \quad \text{when C-peptide data is not available} \tag{2.1f}$$

$$P(t) = \min\left(d_2 P_2, P_{max}\right) + P_N(t). \tag{2.1g}$$

Equation (2.1a) is the independent insulin glucose removal, with G being the total blood glucose concentration (mmol/L). The whole-body insulin sensitivity $S_I(t)$ gives insight into the metabolic condition of the patient and is a time-varying dynamic. The insulin pharamacodynamics are shown in Equations (2.1b) and (2.1c), with Q and I being the interstitial insulin concentration (mU/L) and the plasma insulin concentration (mU/L) respectively. The exogenous insulin input is $u_{ex}(t)$ (mU/min), and the endogenous insulin production $u_{en}(t)$ (mU/min) can be measured from C-Peptide, but if measurements are not feasible, the kinetics can model as such in Equation (2.1f). The gastric absorption of glucose is defined in Equations (2.1d) to (2.1g). The dynamics P_1 and P_2 describe the glucose in the stomach and the gut (mmol), with $P(t)$ representing the glucose appearance in the blood (mmol/min). Any enteral food intake given to the patient is accounted for by $D(t)$ (mmol/min), and additional parenteral glucose is described by $P_N(t)$ (mmol/min).

The ICING model also involves several parameters alongside the previously defined dynamics, which are detailed in Table 2.1. The majority of the patient model parameters are assigned as general population constants observed from clinical trials, and are validated in relevant insulin-glucose control parameter sensitivity studies [18] [15] [21]. The identification of the remaining parameters was estimated with a data-driven approach by fitting the patient parameters to the observed dynamics in 173 patients, and by assessing the impact of prediction errors [21]. Estimating the endogenous glucose removal p_G and the suppression of endogenous glucose produce E_G constants involved assigning all other patient parameters and the insulin sensitivity to constants, then starting a grid search that covers $p_G = 0.001, 0.002, \ldots, 0.1$(min^{-1}) and $E_G = 0.0, 0.1, \ldots, 3.5$ (mmol/min). The joint p_G and E_G value that achieved the lowest fitting and prediction error for blood glucose in Equation (2.1a) was chosen. Ultimately, that was $p_G = 0.006$ (min^{-1}) and $E_G = 1.16$ (mmol/min), while both falling in the ranges observed physiologically. The glucose distribution rate between the stomach and the gut n_I, and cellular insulin clearance rate in the interstitium were estimated similarly by fitting to Q in the Equation (2.1b), a grid search covers a range of $n_I = n_C = 0.0001, \ldots, 0.02$ (min^{-1}), with a value of $n_I = 0.003$ (min^{-1}) resulting in the best predictive performance.

It should be noted that in the ICING model, the nonlinear dynamics are defined in continuous-time. Discretization of the model is necessary as in the following section, the estimation methods presented operate within discrete time, noted as k. If we define the discrete time step as $t_k = t_{k-1} + \Delta t$, an example state $F(t)$ can be discretized according to the Euler method

$$\begin{aligned} \dot{F} &\approx \frac{F(t_k) - F(t_k - \Delta t)}{\Delta t} \\ F_k &\approx F_{k-1} + \Delta t \dot{F}. \end{aligned} \tag{2.2}$$

The error of this numerical approximation of differential equations is bounded by the step size Δt. As a result, larger step sizes will introduce greater approximation errors. Thus, a relatively small step size is necessary to prevent errors from propagating in the estimates.

A probabilistic representation of the ICING model will be utilized in this

Table 2.1: Assigned parameters in the ICING model. The parameters shaded in gray are derived from fitting methods.

Parameter	Notation	Value
Patient endogenous glucose removal	p_G (min^{-1})	0.006
Saturation of insulin-stimulated glucose removal	α_G (L/mU)	0.0154
Suppression of endogenous glucose production	E_G (mmol/min)	1.16
Central nervous system uptake	C_G (mmol/L)	0.3
Glucose distribution volume	V_G (L)	13.3
Glucose distribution rate between the stomach and the gut	n_I (min^{-1})	0.003
Cellular insulin clearance rate in the interstitium	n_C (min^{-1})	$= n_I$
Kidney clearance rate of insulin from plasma	n_K (min^{-1})	0.0542
Liver clearance rate of insulin from plasma	n_L (min^{-1})	0.01578
Saturation of plasma insulin disappearance	α_I (L/mU)	0.0017
Insulin distribution volume	V_I (L)	3.15
Fraction of endogenous insulin hepatic extraction	x_L	0.67
Maximal glucose out flux from the gut	P_{max} (mmol/min)	6.11
Base endogenous insulin production rate	k_1 (mU/min)	45.7
Transport rates between the stomach and the gut	d_1 (min^{-1})	0.0347
	d_2 (min^{-1})	0.0069
Exponential suppression constants	k_2	1.5
	k_3	1000

thesis for Bayesian filtering. The states in Equations (2.1a) to (2.1g), will be described as a Bayesian inference from the previous states and inputs. Modeling the ICING equations as a random process will be done to model the uncertainty of the states. The specific conditional probability distribution will be dependent on the type of Bayesian filter, as seen in the next section.

2.3 State Estimation

Within the scope of this thesis, discrete-time dynamical systems will be modeled in the state-space representation. A state-space model consists of system variables that describe the state of the system, inputs, and measurements. The states are typically formulated as linear or nonlinear equations. A probabilistic representation of the discrete-time state-space can be represented as such

$$\begin{aligned} x_k &\sim p(x_k \mid x_{k-1},\, u_{k-1}) \\ y_k &\sim p(y_k \mid x_k), \end{aligned} \tag{2.3}$$

where $p(x_k \mid x_{k-1},\, u_k)$ and $p(y_k \mid x_k)$ are the stochastic discrete-time dynamic and measurement models, with $x_k \in \mathbb{R}^n$ representing the states, $u_k \in \mathbb{R}^p$ are the inputs, and $y_k \in \mathbb{R}^m$ are the measurements.

Any model has intrinsic approximation errors, and the errors should be taken into account. The approximation error for the states can be modeled as a random process, and artificial process noise can be added into x_k at each time step. This step is important to accommodate errors due to unmodeled dynamics. Choices in the specifics regarding the modeling of process noise of insulin-glucose states will be discussed in Chapter 3.

The objective of state estimators can described as estimating the distribution of the state x_k , given all the measurements up to the current sample $p(x_k \mid y_{1:k})$. In the case of assuming the model is linear and Gaussian, the states and measurements can be written as the difference equations

$$\begin{aligned} x_k &= A_{k-1}x_{k-1} + B_{k-1}u_{k-1} + q_{k-1} \\ y_k &= Hx_k + r_k, \end{aligned} \tag{2.4}$$

where $q_{k-1} \sim \mathcal{N}(0,\, Q_{k-1})$ is the process noise, $r_k \sim \mathcal{N}(0,\, R_k)$ is the measurement noise, we can rewrite the Equation (2.3) as the following

$$\begin{aligned} p(x_k \mid x_{k-1}, u_{k-1}) &= \mathcal{N}(x_k \mid A_{k-1}x_{k-1} + B_{k-1}u_{k-1},\, Q_{k-1}) \\ p(y_k \mid x_k) &= \mathcal{N}(y_k \mid H_k x_k,\, R_k). \end{aligned} \tag{2.5}$$

The closed-form solution of $p(x_k \mid y_{1:k})$ for linear and Gaussian dynamic and measurement functions is known as the Kalman filter, with a full derivation available from Särkkä and Svensson [22]. The filter contains two key parts: the *prediction step* and the *update step*.

1. *Prediction step*

$$\begin{aligned} m_k^- &= A_{k-1}x_{k-1} + B_{k-1}u_{k-1} \\ P_k^- &= A_{k-1}P_{k-1}A_{k-1}^{\mathsf{T}} + Q_{k-1}. \end{aligned} \tag{2.6}$$

2. *Update step*

$$
\begin{aligned}
v_k &= y_k - H_k m_k^- \\
S_k &= H_k P_k^- H_k^\mathsf{T} + R_k \\
K_k &= P_k^- H_k^\mathsf{T} S_k^{-1} \\
m_k &= m_k^- + K_k v_k \\
P_k &= P_k^- - K_k S_k K_k^\mathsf{T}.
\end{aligned}
\tag{2.7}
$$

The Kalman filter is recursive, as the filter starts with an estimate of the initial distribution of the state m_0^- and the covariance P_0^-, and iterates sequentially over discrete time steps k. In practice, the Kalman filter is often a great starting point for an estimator because the assumptions of linearity and Gaussianity often hold in many contexts, but as Equations (2.1a) to (2.1e) are nonlinear, other methods must be considered.

2.3.1 Extended Kalman filter

The extended Kalman filter can accommodate the nonlinearities in the ICING model by approximating the dynamic and measurement models as a linear transformation of the states, inputs, and measurements. The structure of the extended Kalman filter remains the same as the Equations (2.6) and (2.7), but changes are seen in the matrices of the dynamic and measurement models $f(\cdot)$ and $h(\cdot)$. Assuming that the dynamic and measurement models are Gaussian, the prediction and update steps [22] can be defined as

1. *Prediction step*

$$
\begin{aligned}
m_k^- &= f(m_{k-1}, u_{k-1}) \\
P_k^- &= F_x(m_{k-1}) P_{k-1} F_x^\mathsf{T}(m_{k-1}) + F_q(m_{k-1}) Q_{k-1} F_q^\mathsf{T}(m_{k-1})
\end{aligned}
\tag{2.8}
$$

2. *Update step*

$$
\begin{aligned}
v_k &= y_k - h(m_k^-) \\
S_k &= H_x(m_k^-) P_k^- H_x^\mathsf{T}(m_k^-) + H_r(m_k^-) R_k H_r^\mathsf{T}(m_k^-) \\
K_k &= P_k^- H_k^\mathsf{T}(m_k^-) S_k^{-1} \\
m_k &= m_k^- + K_k v_k \\
P_k &= P_k^- - K_k S_k K_k^\mathsf{T},
\end{aligned}
\tag{2.9}
$$

the Jacobian matrices $F_x(m_k^-)$, $F_q(m_k^-)$, $H_x(m_k^-)$, and $H_r(m_k^-)$ are the Taylor series linear approximation of the dynamic and measurement models

$$\begin{aligned}
[F_x(m)]_{j,j'} &= \left.\frac{\partial f_j(x,q)}{\partial x_{j'}}\right|_{x=m_{k-1},\, q=0} \\
[F_q(m)]_{j,j'} &= \left.\frac{\partial f_j(x,q)}{\partial q_{j'}}\right|_{x=m_{k-1},\, q=0} \\
[H_x(m)]_{j,j'} &= \left.\frac{\partial h_j(x,r)}{\partial x_{j'}}\right|_{x=m_k^-,\, r=0} \\
[H_r(m)]_{j,j'} &= \left.\frac{\partial h_j(x,r)}{\partial r_{j'}}\right|_{x=m_k^-,\, r=0}.
\end{aligned} \tag{2.10}$$

The intrinsic bottleneck of the extended Kalman filter can be ascribed to the Gaussian formulation of the state update equations. Any deviations from Gaussianity in the posterior will directly degrade the estimation capabilities of the estimator. Another type of state estimation method is the particle filter, which can accommodate nonlinear and nongaussian dynamic and measurement models by using point estimates from a Monte Carlo approximation of the posterior $p(x_k \mid y_{1:k})$.

2.3.2 Bootstrap particle filter

For state-space modeling, the dynamic and measurement models are run sequentially such that x_k and y_k are represented as a Markov Chain seen in Equation (2.3). In the Kalman filter case, such dynamic and measurement models were assumed to be linear and Gaussian. As a result, the posterior of the state $p(x_k \mid y_k)$ was derived as a linear transformation of the mean and covariance. For the bootstrap particle filter, the posterior of the state is formulated as a Monte Carlo approximation, with $x_k^{(i)}$ denoting a randomly sampled state for N total particles

$$p(x_k \mid y_k) \approx \frac{\sum_{i=1}^{N} p(y_k \mid x_k^{(i)}) p(x_k^{(i)})}{\sum_{j=1}^{N} p(y_k \mid x_k^{(j)})}. \tag{2.11}$$

The steps to implement to formulate the bootstrap particle fiter iteratively can be summarized in the following algorithm [22] [23] [24]

1. Sample N particles from the dynamic function

$$x_k^{(i)} \sim p(x_k \mid x_{k-1}^{(i)},\, u_{k-1}) \quad i = 1, \ldots, N \tag{2.12}$$

2. Update weights from the measurement

$$\begin{aligned} w_k^{*(i)} &\propto p(y_k \mid x_k^{(i)}) \quad i = 1, \dots, N \\ w_k^{(i)} &= \frac{w_k^{*(i)}}{\sum_{j=1}^{N} w_k^{*(j)}} \quad i = 1, \dots, N \end{aligned} \tag{2.13}$$

3. Systematically resample particles from a distribution proportional to the weights

$$\begin{aligned} u_1 &\sim \mathbb{U}\left[0, \frac{1}{N}\right] \\ u_r &= u_1 + \frac{r-1}{N} \qquad r = 2, \dots, N \\ x_k^{(i)} &= \left\{ x_k^{(r)} \quad \text{if } \sum_{j=1}^{i-1} w_k^{(j)} < u_r \leq \sum_{j=1}^{i} w_k^{(j)} \qquad i = 1, \dots, N \right. \\ w_k^{(i)} &= \frac{1}{N} \qquad i = 1, \dots, N \end{aligned} \tag{2.14}$$

2.3.3 Observability in Nonlinear Discrete-Time State-Space Models

Analyzing observability in linear and nonlinear systems assesses how well the states of the model can be inferred from the observations. The observability gramian W_O has traditionally been used in model reduction of linear dynamic systems, but empirical methods have been adapted to accommodate nonlinear systems [25]. The observability gramian provides insight into the states with the most influence on the output by analyzing the rank condition of the observability gramian matrix, which in return allows for analytically evaluating observability.

To empirically calculate the observability gramian, 2^n unique combinations of the initial state $x_0 \in \mathbb{R}^n$ are generated, with n denoting the numbers of states [20] [25] [26]. The initial condition x_0 will be assigned as perturbations $\delta_i \in \mathbb{R}^n$ around a nominal value $x_{nom} \in \mathbb{R}^n$

$$\begin{aligned} x_{0,i} &= x_{nom} + \delta_i \\ \delta_i &= c \cdot S \cdot t_i, \qquad \text{with } S = \operatorname{diag}(x_{\max}). \end{aligned} \tag{2.15}$$

The scalar constant c corresponds to the magnitude of deviation, $S \in \mathbb{R}^{n,n}$ is the diagonal matrix of the maximum of each state and parameter. The row vector $t_i \in R^n$ is the specified perturbation i from the orthonormal matrix

$$
\begin{aligned}
T &= \begin{bmatrix} t_1 & t_2 & \dots & t_{2^n} \end{bmatrix} \in R^{n,2^n} \\
t_1^{\mathsf{T}} &= \frac{1}{\sqrt{2^n}} \begin{bmatrix} -1 & -1 & \dots & -1 \end{bmatrix} \\
t_2^{\mathsf{T}} &= \frac{1}{\sqrt{2^n}} \begin{bmatrix} -1 & -1 & \dots & 1 \end{bmatrix} \\
&\vdots \\
t_{2^n}^{\mathsf{T}} &= \frac{1}{\sqrt{2^n}} \begin{bmatrix} 1 & 1 & \dots & 1 \end{bmatrix}.
\end{aligned}
\tag{2.16}
$$

After calculating the 2^n perturbations of the initial state, the output $y_{1:K,i}$ can be derived sequentially with Equations (2.3) from time steps $k = 1, \dots, K$. Then, the empirical observability with discrete time steps $t = k\,dt$ can be expressed as such

$$
W_O = (cS)^{-1} \sum_{k=1}^{K} T\Psi\left[k\right] T^{\mathsf{T}}\, dt (cS)^{-1},
\tag{2.17}
$$

with the positive definite matrix $\Psi\left[k\right]$ being assigned as the following

$$
\Psi_{i,j}\left[k\right] = \left|y_i\left[k\right] - y_0\left[k\right]\right| \left|y_j\left[k\right] - y_0\left[k\right]\right|.
\tag{2.18}
$$

If the states have varying magnitudes of scale, it is recommended [20] to rerun the observability gramian algorithm again with the following transformations

$$
x_s = \sqrt{\operatorname{diag}(W_O)} \qquad\qquad T_s = \operatorname{diag}^{-1}(x_s).
\tag{2.19}
$$

The new state perturbations are assigned as the transformation $\bar{x} = T_s x$. To make inferences about of the observability gramian, one can check the rank condition of the matrix by analyzing the nonzero singular values of W_0. If there exists singular values near zero (e.g. < 0.1), a subset of the dynamics of the system can not be observed from the measurements.

2.3.4 Related Work

Nonlinear state estimation of the ICING model has been investigated in a previous study by Aguirre-Zapata et al. [19]. The study incorporated an additional measurement and dynamic due to their clinical facility having a continuous glucose machine that can measure the patient's interstitial glucose, but the formulation of patient insulin-glucose dynamics otherwise remained the same. In the methodology of the paper, all tests regarding estimation performance are tested on deviations from one synthetic patient, with each deviation being assigned the same exogenous insulin input per deviation, and food intake is generally not considered. Overall, their synthetic patients do not constitute a reasonable patient model in comparison to real patients.

The study includes three state estimators: the extended Kalman filter, the unscented Kalman filter, and the particle filter. The majority of the study uses discrete time steps of 5 minutes. While the dynamics of patients develop relatively slowly, such a large time step can introduce approximation errors in the discretization of the ICING model. Another notable design choice is their particle filter containing 5000 total particles. Further analysis of various sized particle filters would motivate the overall trade-off estimation performance compared to the time complexity. State estimation remained in the scope of dynamics represented in the ICING model and an additional interstitial glucose dynamic. As a result, there exists further development of state estimators to predict unrepresented ICING model parameters as well.

2.4 Parameter Estimation

As previously mentioned, there exists unrepresented ICING model parameters. Additionally, these parameters contain time-varying characteristics. The whole-body insulin sensitivity S_I or the endogenous insulin u_{en} dynamically fluctuates over time depending on the patient's metabolic condition. As a result, naively assigning either as a general population constant can hinder the performance of the estimators. The unknown time-varying parameters of interest will be denoted as θ in preparation for the next section.

2.4.1 State Augmentation

A practical method to handle unknown parameters is to augment them to the existing state variables $\tilde{x}_k = [x_k, \theta_k]^\mathsf{T}$, such that Equation (2.3) can be redefined as

$$\begin{aligned} \tilde{x}_k &\sim p(\tilde{x}_k \mid \tilde{x}_{k-1},\, u_{k-1}) \\ y_k &\sim p(y_k \mid \tilde{x}_k). \end{aligned} \tag{2.20}$$

An important distinction in Equation (2.20) is that the unknown parameters are treated as stochastic variables with singularity in the process noise. This can be problematic as all models often have approximation errors, and this can drive the extended Kalman filter to ultimately diverge in the Gaussian approximation, as well as introduce sample impoverishment in particle filters [22].

One method to compensate for singularity is to also add artificial process noise ϵ_k to the unknown parameters to allow for random walking in the state-space, thus, accounting for errors in the dynamic function

$$\theta_k = \theta_{k-1} + \epsilon_{k-1}, \qquad \text{with } \epsilon_{k-1} \sim \mathcal{N}(0,\, W_{k-1}). \tag{2.21}$$

With this method, you inherently add another unknown parameter W_k, but such a parameter can be tuned with prior knowledge of the parameters θ_k.

It should be noted that for state augmentation, the dimension of the state-space increases by the number of unknown parameters. In the case of the particle filter, the number of particles will need to be adjusted to maintain accuracy. A simple heuristic, instead of analytically deriving the exact dependency, could be to have the base number of particles P increase by a common factor for each dimension d, such that the total number of particles needed N is

$$N = P \times 2^d. \tag{2.22}$$

2.4.2 Related Work

The IPI method is a nonlinear and nonconvex parameter identification method that has been introduced that can estimate patient-specific parameters as a least squares solution [18]. By linearly interpolating between the blood glucose measurements in times $[t_0, t_1]$,

$$\widetilde{G}(t) = \frac{G(t_1) - G(t_0)}{t_1 - t_0}(t - t_0) + G(t_0), \tag{2.23}$$

the Equation (2.1a) can be rearranged as such for a piecewise constant S_I

$$\begin{aligned} \int_{t_0}^{t_1} \dot{G}\; dt \approx -p_G \int_{t_0}^{t_1} \widetilde{G}(t)\; dt - S_I \int_{t_0}^{t_1} \widetilde{G}(t) \frac{Q(t)}{1 + \alpha_G Q(t)}\; dt \\ + \frac{1}{V_G} \int_{t_0}^{t_1} P(t) + E_G - C_G\; dt \\ G(t_1) - G(t_0) \approx -p_G \int_{t_0}^{t_1} \widetilde{G}(t)\; dt - S_I \int_{t_0}^{t_1} \widetilde{G}(t) \frac{Q(t)}{1 + \alpha_G Q(t)}\; dt \end{aligned}$$

$$+\frac{1}{V_G}\int_{t_0}^{t_1} P(t) + E_G - C_G \; dt, \tag{2.24}$$

and S_I can be expressed as a linear equation

$$\underbrace{\int_{t_0}^{t_1} \widetilde{G}(t)\frac{Q(t)}{1+\alpha_G Q(t)}\,dt}_{A(t_0,t_1)}\; S_I \approx \underbrace{\begin{pmatrix} G(t_0) - G(t_1) - p_G \displaystyle\int_{t_0}^{t_1} \widetilde{G}(t)\; dt \\ + \dfrac{1}{V_G}\displaystyle\int_{t_0}^{t_1} (P(t) + E_G - C_G)\; dt \end{pmatrix}}_{b(t_0,t_1)}. \tag{2.25}$$

It is often the case with longer measurement periods $(t_1 - t_0)$, that identifying multiple S_I values within the time period may be necessary. For instance, if every 120 minutes glucose measurements are taken from the patient, but every 60 minutes a patient's insulin sensitivity should be identified $S_I \in \mathbb{R}^2$. Both insulin sensitivity parameters at 60 minutes S_I^{60} and 120 minutes S_I^{120} can be derived as the least squares solution from the system of linear equations

$$\begin{bmatrix} A(t_0, t_0+30) & 0 \\ A(t_0+30, t_0+60) & 0 \\ 0 & A(t_0+60, t_0+90) \\ 0 & A(t_0+90, t_0+120) \end{bmatrix} \begin{bmatrix} S_I^{60} \\ S_I^{120} \end{bmatrix} = \begin{bmatrix} b(t_0, t_0+30) \\ b(t_0+30, t_0+60) \\ b(t_0+60, t_0+90) \\ b(t_0+90, t_0+120) \end{bmatrix}. \tag{2.26}$$

The fundamental flaw of the IPI method is the assumption of linear blood glucose dynamics between measurements of times $[t_0, t_1]$. If we closely analyze the Figure 2.1, linearly interpolating blood glucose introduces residuals, due to fitting errors. Consequently, the residuals will propagate into the integrals in the Equation (2.25), resulting in errors in the parameter calculation. Other methods not relying on linearization of the measurements must be explored to allow for realistic estimation of parameters, thus giving better insight of the patient's health status to clinicians.

2.5 Summary

The ICING model provides a foundation to model glucose removal, insulin pharmacokinetics, and gastric absorption of glucose for tight glucose control in critically ill patients. Utilizing nonlinear state estimators such as the extended Kalman filter and the bootstrap particle filter allows for

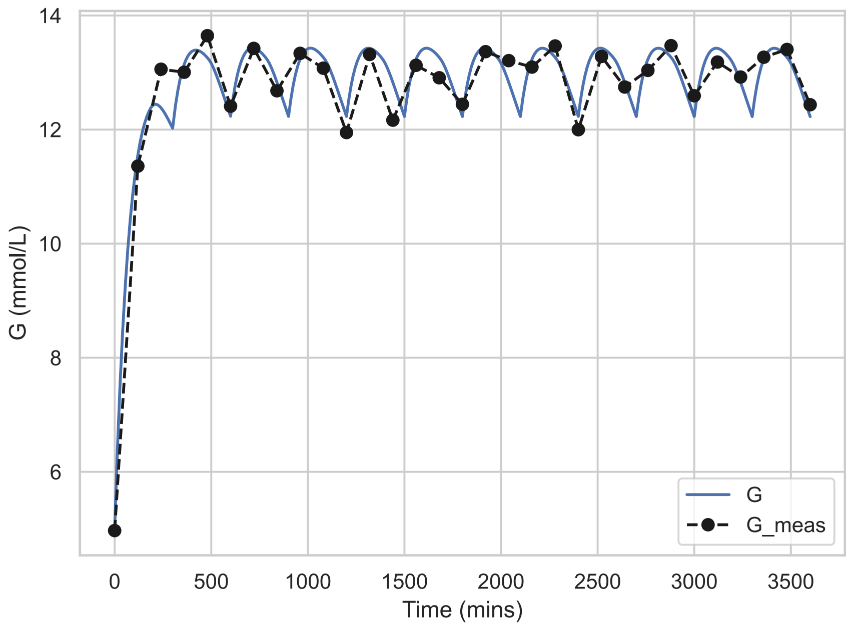

Figure 2.1: Blood glucose and its' corresponding measurements from an example patient trajectory from the ICING model.

optimal estimation of the states over time with consideration of model approximation errors and measurement noise. Consideration of realistic synthetic patients and systematically motivated parameterization of filters will improve generalizability of the estimators. The current method of IPI is formulated with strict assumptions of blood glucose measurements, which limits the ability for clinicians to infer the health of the patient from the approximated patient-specific parameters. More in-depth parameter estimation methods can improve model identifiability for glucoregulation in critically ill patients.

Chapter 3

Methodology

The necessary methodology to achieve the goals introduced in Section 1.4 will be presented in this chapter. It begins with the design of the synthetic patient cohort used for simulation of patients in this paper. The observability of the ICING model is analyzed to learn if the states can be inferred from the measurements. Based on the observed performance, nonlinear state estimators are utilized to assess the model's ability to estimate unmeasured dynamics. Finally, the parameter estimation methods used to estimate unknown constant and time-varying insulin sensitivity are introduced.

3.1 Synthetic Patient Cohort

A synthetic patient cohort will be simulated to model clinical insulin-glucose data observed in patients in the intensive care unit. A patient model consists of simulated insulin-glucose dynamics (G, Q, I, P_1, P_2, u_{en}, P) represented in Equations (2.1a) to (2.1g), known insulin sensitivity S_I, exogenous insulin u_{ex}, enteral feed D, and parenteral feed P_N. The population constants listed in Table 2.1 are assigned to the constant parameters for all synthetic patients. To introduce variance between patients, a synthetic patient is given a random constant input observed from doses given to patients in the Karolinska University Hospital

$$\begin{aligned} u_{ex} &\sim \mathcal{U}[0,\, 166] \text{ mU/min} \\ D &\sim \mathcal{U}[0.2,\, 0.4] \text{ mmol/min} \\ P_N &\sim \mathcal{U}[0.2,\, 0.4] \text{ mmol/min.} \end{aligned} \tag{3.1}$$

The insulin sensitivity has been shown to follow a log-normal distribution [27]. Sampling the insulin sensitivity from a log-normal is important to adhere to the physiological constraints typically observed in patients, as the insulin sensitivity is small in magnitude, and sampling from a normal distribution could shift the insulin sensitivity negative or too large in magnitude, which can possibly result in poor estimation. The values chosen within this study will be drawn from the distribution*

$$S_I = \exp(X), \qquad \text{with } X \sim \mathcal{N}(-8.05,\, 0.70^2). \tag{3.2}$$

To simulate patient data given the inputs and parameters, each synthetic patient needs to be assigned a unique initial state† for the filter

$$x_0 = \begin{bmatrix} G_0 & I_0 & Q_0 & P_{1,0} & P_{2,0} & u_{en,0} \end{bmatrix}. \tag{3.3}$$

If it assumed that the patient begins at a steady state, it is possible to derive the initial states by solving for the roots of the differential equations (2.1a) to (2.1f). This can be computed with the python solver function `scipy.optimize.fsolve`‡ for each patient in the cohort.

Given all the initial states, known insulin sensitivity, and the patient inputs, the patient cohort can be generated by solving the differential equations (2.1a) to (2.1e) with a time step of 1 minute for each patient. The results will be inputs to Equations (2.1f) and (2.1g). The ODE solver `scipy.integrate.solve_ivp`§ function in python will be applied with the Explicit Runge-Kutta of order 5(4) integration method [28] to solve for the necessary dynamics.

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*These parameters are approximated according to previous work from Davidson et al. [27]

†The glucose appearance in the blood P is not included as an initial state in the state-space model. This is a result of patients not reaching the maximum glucose out flux from the gut P_{max} at the beginning of the simulation. Thus, it simplifies to a deterministic equation.

‡https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.fsolve.html

§https://docs.scipy.org/doc/scipy/reference/generated/scipy.integrate.solve_ivp.html

3.2 Observability of the Intensive Control Insulin-Nutrition-Glucose Model

In Section 3.1, synthetic patients were created by first randomly sampling initial dynamics, and then utilizing an ODE solver to solve the dynamics according to the ICING model. For the ICING model to be classified as observable, it must be possible to infer the insulin-glucose states, given a sufficient number of observations. To establish if the formulation of the ICING model is observable, a mathematical analysis can determine the relationship of the insulin-glucose states and the observations as previously discussed in Section 2.3.3.

1000 patients will be randomly sampled to investigate the observability of the ICING model across various permutations of each random nominal state vector. The maximum value of each state* x_{max} will be chosen as the maximum values of the patient cohort states found in the previous section 3.1

$$S = \mathrm{diag}(x_{max}), \tag{3.4}$$

as well with a maximum perturbation constant of $c = 0.1$. With the following defined inputs, the observability gramian W_O can be derived to analyze the singular values of all the calculated matrices using single value decomposition. To also analyze which states are the most observable respectively to each other, the diagonals of the observability gramian can be used to propose that diagonal entries greater in magnitude provide more information in relation to the observations. Formulating an approximation of the observability will provide insight into the performance of the estimators in the following section.

3.3 Nonlinear State Estimation

The extended Kalman filter and the particle filter are two different nonlinear state estimation methods presented in Section 2.3. To compare each estimator with the ICING model, the computational complexity $\mathcal{C}$ of estimating all states over the total time and the MSE of the blood glucose concentration estimates will be recorded for all patients and filters in the given synthetic patient cohort. The simulations will span 12 hours to allow each estimator to converge, as well as a discrete time step of 1 minute, with glucose measurements corrupted with

*For this calculation, u_{en} and P are omitted. This is due to the fact that these dynamics are deterministic at $t = 0$.

additive Gaussian noise ($\mathcal{N}(0, 0.125)$) sampled every 120 minutes. There will be offsets in the initial states of $\pm 2\%, \pm 10\%, \pm 25\%$, with each offset having a unique synthetic cohort of 500 patients. Adding an offset to the initial state will gauge how well each estimator can adapt to various initial perturbations in the estimate.

Artificial process noise will be added to the state estimates for both the extended Kalman filter and the particle filter. The process noise q will be sampled at each discrete time step for both the extended Kalman filter and the particle filter roughly according to the following normal distribution

$$\begin{aligned} q_k &\sim \mathcal{N}\big(0, \operatorname{diag}(\boldsymbol{\sigma})\big) \\ \boldsymbol{\sigma} &= \frac{1}{120}\left[\left(\tfrac{\operatorname{med}(G)}{100}\right)^2 \quad \left(\tfrac{\operatorname{med}(I)}{100}\right)^2 \quad \cdots \quad \left(\tfrac{\operatorname{med}(P_2)}{100}\right)^2 \quad 1.0\times 10^{-8}\right]^{\mathsf{T}}. \end{aligned} \tag{3.5}$$

The variance for the dynamics G, I, Q, P_1, P_2 are chosen as the scaled median for the population statistics found in Section 3.1. A 1% deviation is chosen as the dynamics in the synthetic patients are relatively slow between measurements, and do not deviate much relative to the state. The process noise for the enteral insulin is assigned as a relatively small value in comparison to the other states to have the insulin sensitivity absorb more of the uncertainty of the glucose distribution between the gut and the stomach, instead of the endogenous insulin.

The extended Kalman filter parameters will be defined as

1. Initial covariance matrix of

$$P_0 = \operatorname{diag}\left(\log_{10}\left[\operatorname{med}(G) \quad \operatorname{med}(I) \quad \cdots \quad \operatorname{med}(u_{en})\right]^{\mathsf{T}}\right) \tag{3.6}$$

2. Process noise covariance matrix of

$$Q = \operatorname{diag}(\boldsymbol{\sigma}) \tag{3.7}$$

3. Measurement noise covariance matrix of

$$R = \big[0.125\big] \tag{3.8}$$

Particle filters containing $(100,\ 200,\ 300,\ 400,\ 500,\ 750,\ 1000,\ 1500,\ 2000)$ particles will be used to compare the relationship of the estimation performance over an increasing number of particles. The notation of the extended Kalman filter will be abbreviated as EKF and for a particle filter with

N particles, it will be represented as PF N. The particle filter will be initialized by sampling particles from a multivariate Gaussian distribution with a mean of the initial state x_0, and a covariance equal to P_0.

$$x_0^{(i)} \sim \mathcal{N}(x_0, P_0) \qquad i = 1, \ldots, N. \tag{3.9}$$

To define the computational complexity of the estimators given the experimental setup, let τ denote the average total time to run each particle filter and the extended Kalman filter in seconds

$$\begin{aligned} \tau_{\text{min}} &= \min(\tau) \\ \mathcal{C}_i &= \frac{\tau_i}{\tau_{\text{min}}}. \end{aligned} \tag{3.10}$$

The total time complexity is relative to the machine running the particle simulation. Hence, using a normalized scale for τ maintains repeatability across different environments, as each particle filter is designed in software such that the time complexity is linearly dependent on the amount of particles. The error of the estimators will be represented as the MSE of the true blood glucose concentration sequence G and an estimated blood glucose concentration sequence $\hat{G}$ of length L

$$\text{MSE}_i = \frac{1}{L}\sum_{j=1}^{L}\left(G_i[j] - \hat{G}_i[j]\right)^2 \qquad i = 1, \ldots, K. \tag{3.11}$$

A Pareto curve can be utilized to visualize the "optimal" number of particles in terms of the trade-off between computational complexity and error. The resulting number of particles needed per dimension will be calculated to gauge the rough number of particles needed to include additional parameters in the state-space in the following section.

3.4 Parameter Estimation

The IPI method introduced in Section 2.4.2 will be the baseline implementation of estimating insulin sensitivity in patients and will be compared against the extended Kalman filter and the particle filter with state augmentation. Each method will try to estimate a constant insulin sensitivity and a piecewise insulin sensitivity that steps to a new value twice throughout the experiment to draw conclusions on the limitations of each estimator. The simulation will last for 60 hours, as changes in insulin sensitivity are relatively slow in time, with discrete time steps of 1 minute. There will be 500 total patients, and each filter will be assigned an initial state of a $\pm 10\%$ deviation from the true initial state. At 20 and 40 hours, the piecewise insulin sensitivity will be randomly assigned a new value $S_{I,1}$ represented from the current value $S_{I,0}$

$$\begin{aligned} z &\sim \mathbb{U}[0.75,\, 0.95] \\ S_{I,1} &= \begin{cases} S_{I,0}(1-z), & \text{with probability } 0.5, \\ S_{I,0}(1+z), & \text{with probability } 0.5. \end{cases} \end{aligned} \tag{3.12}$$

To ensure the insulin sensitivity is constrained to a physiologically relevant value, $S_{I,1}$ will be clamped to the minimum and maximum insulin sensitivity observed from Equation 3.2. Due to an additional state being included in the state-space, the process noise in Equation (3.5) needs to be appended to consider the uncertainty of S_I. As previously mentioned in Section 3.1, insulin sensitivity has been observed to be log-normal, which would translate to process noise that is multiplicative, not additive. Multiplicative noise follows a similar structure to Equation (3.12), such that

$$\begin{aligned} S_{I,k} &= f(S_{I,k-1},\, u_{k-1})\, q_{S_I,k-1}, \qquad \text{with } q_{S_I,k-1} \sim \mathcal{N}(1,\, \sigma^2_{S_I}) \\ &= S_{I,k-1}\, q_{S_I,k-1}. \end{aligned} \tag{3.13}$$

The multiplicative noise variance $\sigma^2_{S_I}$ will be chosen for the desired coefficient of variation CV for a constant and time-varying insulin sensitivity

$$\begin{aligned} CV &= \frac{\sigma_{S_I}}{\mu} \\ \sigma_{S_I} &= CV. \end{aligned} \tag{3.14}$$

Each estimate in the extended Kalman filter and the particle filter will include sampling from the additive process noise distribution from Equation (3.5), but the insulin sensitivity estimate is multiplied with a sampled random variable in Equation (3.13). The extended Kalman filter configuration for this section is the following

1. Initial covariance matrix of

$$\hat{P}_0 = \operatorname{diag}\left(\log_{10}\left[\operatorname{med}(G) \quad \cdots \quad \operatorname{med}(u_{en}) \quad \max(S_I)^2\right]^{\mathsf{T}}\right) \tag{3.15}$$

2. Process noise matrix of

$$\hat{Q} = \text{diag}\left(\begin{bmatrix} \boldsymbol{\sigma} \\ \frac{\max(S_I)^2}{120} \end{bmatrix}\right) \tag{3.16}$$

3. Measurement noise of R

The initial particles in the particle filter will be sampled with the updated initial covariance matrix

$$x_0^{(i)} \sim \mathcal{N}(x_0,\, \hat{P}_0). \tag{3.17}$$

The particle filter will contain the extrapolated "optimal" number of particles chosen from the previous section to accommodate another dimension in the state-space. For a constant insulin sensitivity, the coefficient of variation can be assigned a smaller magnitude, as the estimator doesn't need to accommodate for larger deviations in the estimate.

Assessing the estimation of the insulin sensitivity will be evaluated using the MSE of the true insulin sensitivity S_I and the estimated insulin sensitivity $\hat{S}_I$ for a given patient i, and a insulin sensitivity measurement of length L

$$\text{MSE}_{S_I,i} = \frac{1}{L}\sum_{j=1}^{L}\Big(S_{Ii}[j] - \hat{S}_{Ii}[j]\Big)^2. \tag{3.18}$$

In the next chapter, the results of the synthetic patient cohort generation, observability, state estimation, and parameter estimation will be presented and analyzed.

Chapter 4

Results

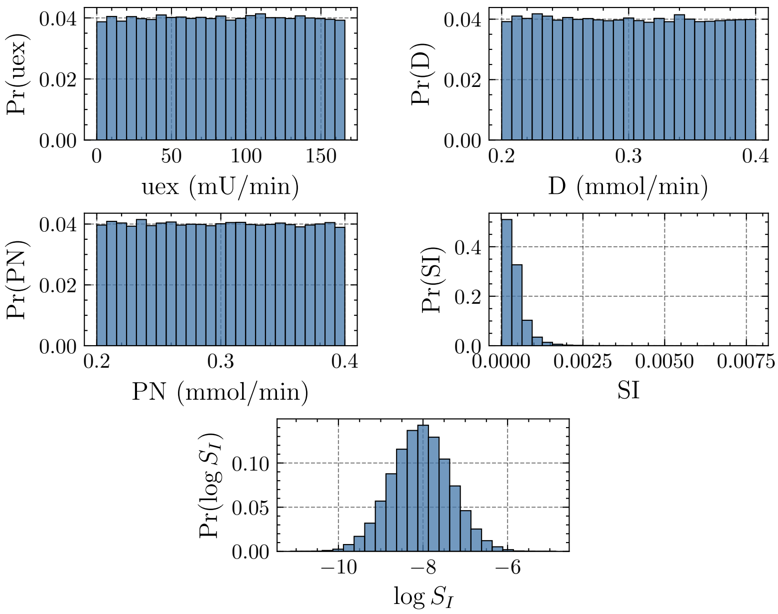
4.1 Synthetic Patient Cohort

Approximating probability mass functions was used to analyze the characteristics of the target population of critically ill patients in the Karolinska University Hospital. Figure 4.1 is divided into two subfigures, with Figure 4.1a consisting of the histograms of the inputs u_{ex}, D, and P_N, as well as the insulin sensitivity parameter S_I, and Figure 4.1b is visualizing the histograms of the derived initial states G, I, Q, P_1, P_2, P, u_{en}.

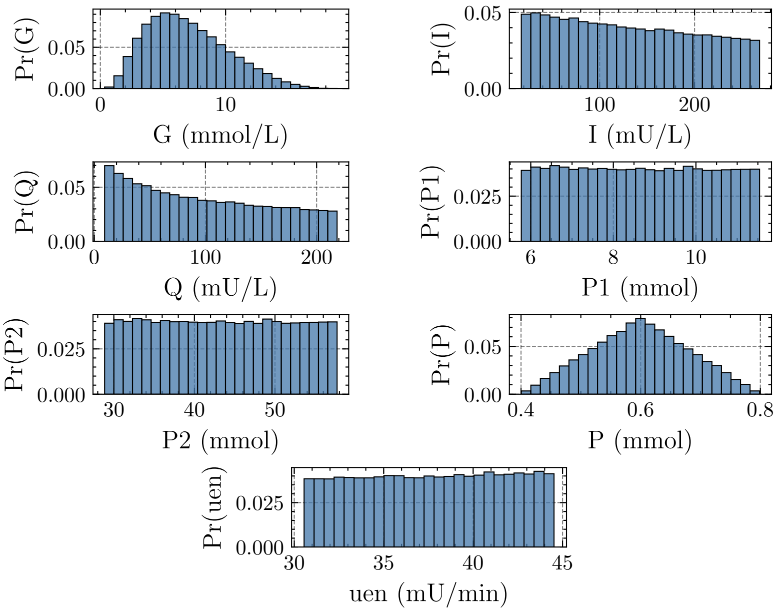
Sampling numerous synthetic patients presented a diverse distribution of realistic initial states from the randomly assigned inputs and insulin sensitivity. This allowed for sampling of many unique variations of patient insulin-glucose states over time without relying on manual interventions. Previous work in synthetic patient cohorts in glycaemic control was often limited to perturbations in previously recorded clinical data [16] or from integral-fitting [17] [18], but in this approach, a clear computational method towards reproducible generation of patient data was made feasible to analyze and compare the methods in the following sections.

4.2 Observability of the Intensive Glucose Insulin-Nutrition-Glucose Model

To investigate the observability of the ICING model, two methods were investigated. In Figure 4.2a, a histogram of eigenvalues from the 1000 calculated empirical observability gramians were plotted, and were transformed to a log-scale to easier visualize the spread of values. The red

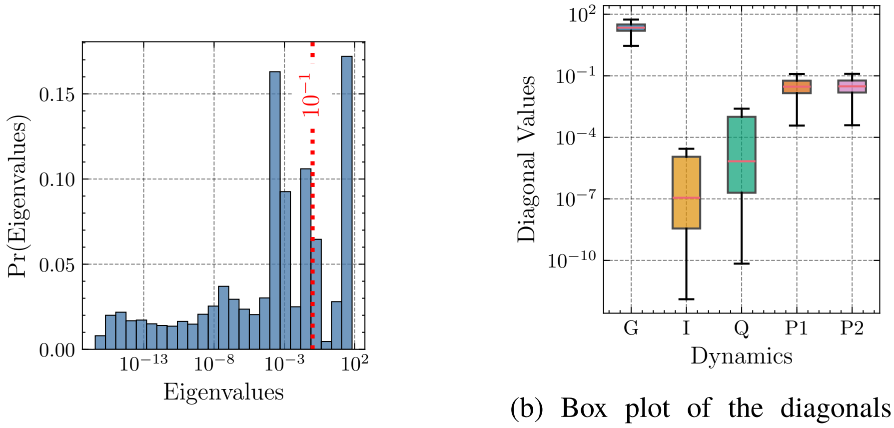


(a) Inputs and parameters



(b) Initial States

Figure 4.1: Histograms of characteristics of the synthetic patient cohort.



(a) Histogram of the eigenvalues

(b) Box plot of the diagonals across dynamics

Figure 4.2: Analysis across all calculated empirical observability gramians.

dotted line denotes the threshold chosen to distinguish the eigenvalues that are considered observable (> 0.10) or unobservable (≤ 0.10). The eigenvalues range from values of 10^{-15} to 10^{2}, with the number of states with eigenvalues considered observable is approximately 2, which would suggest that about a quarter of the states provide insight into the observations. In the other Figure 4.2b, a box plot for each dynamic in the empirical observability gramian is plotted to show the respective locality of the diagonals $\mathrm{diag}(W_O)$ over all patients. As a result of scaling all dynamics in the empirical observability gramian, the diagonal values are comparable amongst the dynamics.

The blood glucose concentration G is the most observable. This is to be expected as the observations in our state-space model are noisy blood glucose measurements. The blood glucose appearance in the stomach and the gut are the next observable states, with a median roughly of 2.95×10^{-2} and 3.07×10^{-2}, as well as an approximate interquartile range of 4.33×10^{-2} and 4.35×10^{-2} respectively. In Equation (2.1a), the change in G is dependent on a constant consisting of a scaled multiple of the total glucose appearance P, which is a function of both the glucose appearance in the stomach and gut P_1, P_2. Thus, this representation can be attributed to why P_1 and P_2 are more observable compared to the plasma insulin I and the interstitial insulin Q, as the fraction $\frac{Q}{1+\alpha_G Q}$ is being multiplied by a relatively small value S_I, and changes in I take time to propagate to Equation (2.1a).

Overall, the ICING model can not be classified as observable. In each observation, only the blood glucose and possibly the total glucose appearance can be uniquely identified. In practice, you could add more observations to increase the observability, but in the context of insulin-glucose dynamics, a lot of institutions only have access to blood glucose measurements, not plasma insulin or C-peptide data. While poor observability is a limitation in state-space models, the following sections show that the model has capabilities, albeit imperfect, to estimate insulin-glucose states.

4.3 Nonlinear State Estimation

In the first stage of comparing different nonlinear state estimators, each estimator is given an initial state that has a deviation in each state from the true initial state. The MSE is averaged among all patients for each deviation in Table 4.1, as well as the standard deviation SD of the error, and an average MSE across all deviations is calculated. The distribution of states from Section 4.1 allows for the corresponding initial and process noise covariance matrices

$$\begin{aligned} P_0 &= \operatorname{diag}\left(\begin{bmatrix}1.00 & 3.00 & 3.00 & 1.00 & 2.00 & 1.00\end{bmatrix}^\mathsf{T}\right) \\ Q &= \frac{1}{120}\operatorname{diag}\left(\begin{bmatrix}0.005 & 1.00 & 1.00 & 0.005 & 0.025 & 1.00\times 10^{-8}\end{bmatrix}^\mathsf{T}\right). \end{aligned} \tag{4.1}$$

The extended Kalman filter overall performs the worst compared to the particle filters. When there are large deviations in the initial state of $\pm 25\%$, the MSE (0.265, 0.117) and the standard deviation (0.85, 0.55) are large respectively in comparison to a particle filter with only 100 particles (MSEs of 0.057 and 0.038, SDs of 0.17 and 0.12). Improvements in performance compared to the extended Kalman filter can be regarded in a particle filter with just 100 particles, highlighting the particle filter's ability to adapt quickly to errors in the state estimates when resampling after the initial measurement. In the deviations of $\pm 2\%$ and $+10\%$, the extended Kalman filter outperforms or is comparable to the particle filters. It is a possibility that when resampling happens at the beginning of the simulation, regardless of the deviation, that the particle filter may be over-correcting due to the initial error, till it adjusts after receiving another measurement.

To compare the overall performance of the filters, it is helpful to analyze Figure 4.3. The x-axis is the computational complexity of each estimator. The minimum time to run amongst all the estimators was the extended filter. Thus, its computational complexity is 1. The computational complexity of the rest of the estimators is the multiple of time to run relative to the extended Kalman filter. The y-axis is the average MSE for all deviations of the initial states, plotted in the log-domain. In terms of error, there is a plateau in improvements

Table 4.1: Mean MSE for different estimators and deviations of the initial state. Standard deviations of the MSE are shown in parentheses.

Estimator	**Deviations**						**Avg**
	-25%	-10%	-2%	+2%	+10%	+25%	
EKF	0.265	0.087	0.040	0.028	0.030	0.117	0.095
	(0.85)	(0.23)	(0.11)	(0.08)	(0.10)	(0.55)	
PF 100	0.057	0.031	0.035	0.037	0.035	0.038	0.039
	(0.17)	(0.08)	(0.09)	(0.10)	(0.10)	(0.12)	
PF 200	0.055	0.033	0.035	0.036	0.035	0.037	0.038
	(0.16)	(0.08)	(0.09)	(0.10)	(0.10)	(0.12)	
PF 300	0.051	0.033	0.035	0.035	0.035	0.036	0.037
	(0.14)	(0.08)	(0.09)	(0.10)	(0.10)	(0.12)	
PF 400	0.046	0.033	0.034	0.035	0.035	0.036	0.037
	(0.13)	(0.08)	(0.09)	(0.10)	(0.10)	(0.11)	
PF 500	0.046	0.033	0.034	0.035	0.035	0.036	0.037
	(0.13)	(0.08)	(0.09)	(0.09)	(0.10)	(0.11)	
PF 750	0.046	0.033	0.034	0.034	0.035	0.035	0.036
	(0.13)	(0.08)	(0.09)	(0.09)	(0.10)	(0.11)	
PF 1000	0.038	0.033	0.034	0.035	0.035	0.036	0.035
	(0.10)	(0.08)	(0.09)	(0.09)	(0.10)	(0.12)	
PF 1500	0.036	0.034	0.034	0.035	0.036	0.036	0.035
	(0.09)	(0.08)	(0.09)	(0.10)	(0.10)	(0.12)	
PF 2000	0.037	0.033	0.035	0.035	0.036	0.037	0.035
	(0.09)	(0.08)	(0.09)	(0.10)	(0.10)	(0.12)	

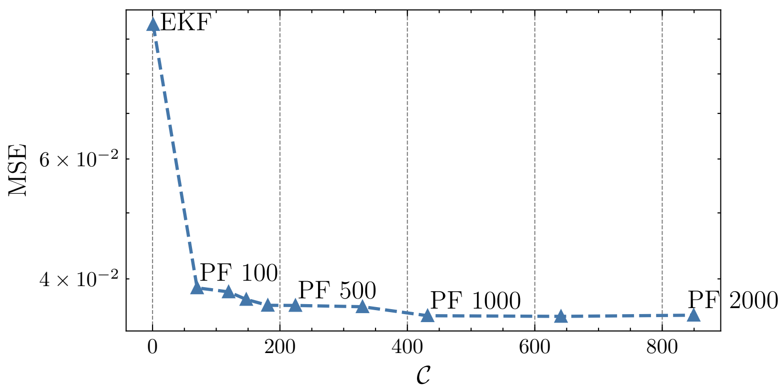


Figure 4.3: Comparison of the total average computational complexity and error across all estimators.

in performance after 750 particles. The "optimum" number of particles was chosen as 750 particles, as there is a balance of moderate performance and lower computational cost. For the next section, the insulin sensitivity will be included in the state-space. Consequently, the state-space increases to 7. To have a rough range of the number of particles needed to accommodate a larger dimension in the state-space, the approximate number of particles 1500 particles according to Section 2.4.1.

The chosen "optimal" number of particles is much less computationally expensive compared to previous work [19], which used 5000 particles for a state-space with the same number of dimensions. Although the Aguirre et al. study did contain nongaussian measurement noise, the number of particles to estimate the posterior distribution with a known measurement model prior likely does not require that many particles, even if an exponential heuristic was chosen for the particle calculation. If we decided to choose 5000 particles, there would likely be little marginal gain in the performance and would result in 5 times the computational complexity. For a system that may require an estimator to run in approximately real-time, an estimator with 5000 particles would require lots of computational resources. Hence, our approach is much more practical in clinical settings. If very limited computational resources are allotted, smaller sized particle filters or the extended Kalman filter can be utilized instead.

4.4 Parameter Estimation

State augmentation of insulin sensitivity in the extended Kalman filter and the particle filter was implemented to compare the estimation performance of the existing state estimators against the IPI method. To accommodate the inclusion of the insulin sensitivity in the state-space, the initial and process noise covariance is appended as such

$$\begin{aligned} P_0 &= \operatorname{diag}(\boldsymbol{\sigma}_{\boldsymbol{P_0}}) \\ Q_{\text{ constant } S_I} &= \frac{1}{120}\operatorname{diag}(\boldsymbol{\sigma}_{\boldsymbol{Q}_{\text{const}}}) \\ Q_{\text{ time-varying } S_I} &= \frac{1}{120}\operatorname{diag}(\boldsymbol{\sigma}_{\boldsymbol{Q}_{\text{tv}}}), \end{aligned} \tag{4.2}$$

$$\begin{aligned} \boldsymbol{\sigma}_{\boldsymbol{P_0}} &= [1.00,\ 3.00,\ 3.00,\ 1.00,\ 2.00,\ 1.00, \\ &\quad\ 1.00 \times 10^{-5}]^{\mathsf{T}} \\ \boldsymbol{\sigma}_{\boldsymbol{Q}_{\text{const}}} &= [0.005,\ 1.00,\ 1.00,\ 0.005,\ 0.025,\ 1.00 \times 10^{-8}, \\ &\quad\ 8.00 \times 10^{-4}]^{\mathsf{T}} \\ \boldsymbol{\sigma}_{\boldsymbol{Q}_{\text{tv}}} &= [0.005,\ 1.00,\ 1.00,\ 0.005,\ 0.025,\ 1.00 \times 10^{-8}, \\ &\quad\ 0.333]^{\mathsf{T}}. \end{aligned} \tag{4.3}$$

Both constant and time-varying insulin-sensitivity were used to investigate how characteristics of the parameter affect estimation. Figures 4.4 and 4.5 display the performance of each method according to the time dependency of the parameter. MSE in these figures depicts the mean error for each time step, not the length of the sequence.

The IPI method in Figure 4.4 illustrates how the estimation accuracy degrades drastically if any measurement noise is present in the calculation. Therefore, the performance is heavily-dependent on the quality of the measurements. Any estimation of the insulin sensitivity from the IPI method does not provide any information on the insulin sensitivity, as the output never converges towards the true value(s) in a practical setting with noisy measurements. For this reason, the IPI method is removed from the comparison of the extended Kalman filter and particle filter in the following section.

In Figure 4.5, both the extended Kalman filter and the particle filter converge towards the true constant insulin sensitivity. The extended Kalman filter on average plateaus in the estimation of the insulin sensitivity around a MSE of 10^{-10}, while the particle filter with 1500 particles on average roughly

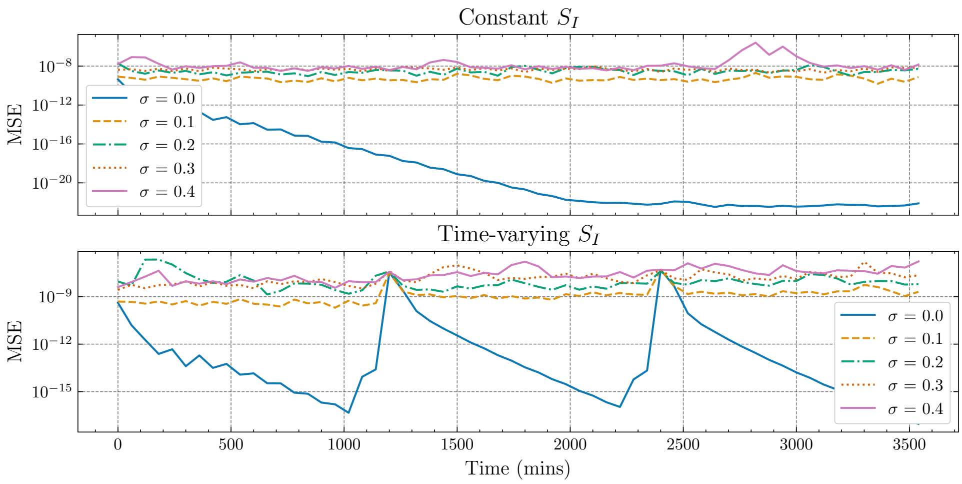


Figure 4.4: Integral-based method insulin sensitivity estimation performance across varying measurement noise variances.

converges closer to the true value with a MSE close to 10^{-11}. The extended Kalman filter often has higher variance in the estimate because particle filters average the variance over all the particles. Thus, the extended Kalman filter can also converge to the true value, but on average performs worse than the particle filter.

In the time-varying parameter estimation case, it can be seen in Figure 4.5 that the extended Kalman filter struggles to converge compared to the particle filter. This can be attributed to the extended Kalman filter's linearization approximation in the dynamic models, as when having a parameter transition to a new value quickly, the mismatch in the Jacobian linearization causes the filter to diverge. About 30% of the extended Kalman filter runs (150 patients) diverged, and had to be discarded entirely. Consequently, the extended Kalman filter is unreliable regarding notable unaccounted for changes in the parameters of interest. The particle filter can be seen to adapt to the steps in the insulin sensitivity at 1200 and 2400 minutes, as the MSE decreases in the minutes following. The particle filter largely remains unaffected by the stationarity of the insulin-sensitivity due to the particle filter's ability to approximate nonlinear and nongaussian dynamic models, unlike the extended Kalman filter. To conclude, the overall performance in parameter estimation is shown in Table 4.2, with the particle filter achieving the lowest MSE in estimating both constant and time-varying insulin sensitivity.

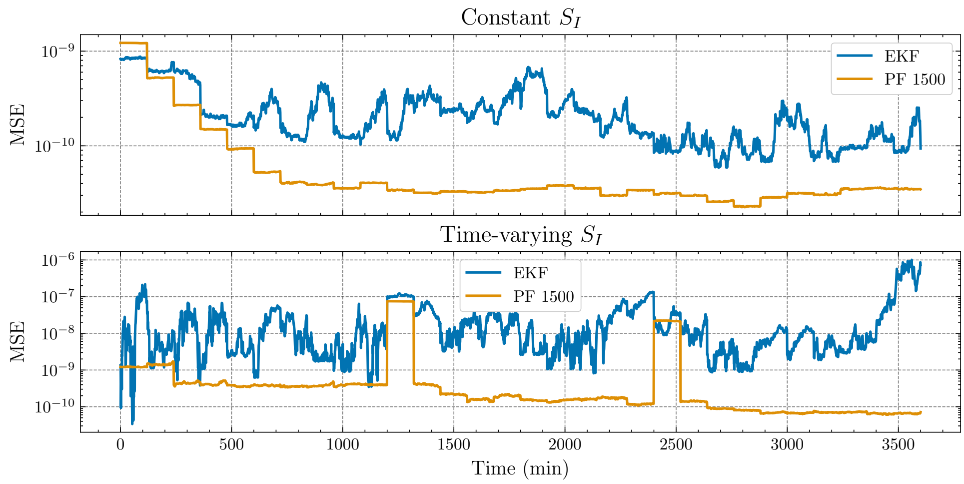


Figure 4.5: Mean MSE over time in parameter estimation methods estimating insulin sensitivity.

(a) Constant S_I

Method	**Deviation**		**Avg**
	-10%	**+10%**	
EKF	9.57×10^{-10} (3.26×10^{-9})	5.03×10^{-9} (2.82×10^{-8})	3.00×10^{-9} (2.02×10^{-8})
PF 1500	7.21×10^{-10} (3.64×10^{-9})	7.02×10^{-10} (3.65×10^{-9})	7.11×10^{-10} (3.65×10^{-9})

(b) Time-varying S_I

Method	**Deviation**		**Avg**
	-10%	**+10%**	
EKF	8.66×10^{-5} (2.61×10^{-3})	9.73×10^{-5} (2.64×10^{-3})	9.19×10^{-5} (2.63×10^{-4})
PF 1500	2.73×10^{-8} (1.82×10^{-7})	2.71×10^{-8} (1.79×10^{-7})	2.72×10^{-8} (1.80×10^{-7})

Table 4.2: Comparison of the mean MSE in parameter estimation methods estimating insulin sensitivity. Standard deviations of the MSE are shown in parentheses.

Chapter 5

Conclusions

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Tight glycaemic control for critically ill patients in the intensive care unit is crucial to prevent adverse outcomes influenced by metabolic fluctuations associated with insulin-resistance. Traditional insulin intensive therapy in patients has been demonstrated to inadequately sustain glucose concentration in a safe and stable range. Utilizing clinically-validated models allows for modeling unknown insulin-glucose dynamics with state estimation methods. Previous work in estimating insulin-glucose dynamics failed to consider a diverse synthetic patient cohort with differing nutrition inputs, as well as motivation for the parameterization of their proposed filters. In this project, a framework to sample synthetic patients was developed to model critically ill patients residing in the Karolinska University Hospital, enabling estimation of insulin-glucose dynamics. The Intensive Control Insulin-Nutrition-Glucose model used in this study and previous work was found to not be observable when computing the empirical observability gramians, further highlighting the limitations of modeling glucoregulation from blood glucose measurements alone. The extended Kalman filter was identified to struggle in estimating blood glucose with large deviations from the true state in the initial prediction ($\pm 10\%$ and $\pm 25\%$) in comparison to the particle filter, but was measured to be run significantly faster. A particle filter with 750 particles achieved the best accuracy-complexity tradeoff, yielding more than a 6 times reduction in complexity from the previous method using 5000 particles. To extend upon previous work, estimation of a parameter not represented in the Intensive Control Insulin-Nutrition-Glucose model formulation was performed. The particle filter managed to estimate both constant and time-varying insulin sensitivity, while the extended Kalman filter only reliably estimated the former. Consequently, drawing attention to the limitations of the linearization approximations seen in the extended Kalman filter. The methodology proposed in this study allows for clinicians to follow a systematic process to develop their own model-based tight glycaemic control method.
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"Keywords[eng]": €€€€
Bayesian filter, Blood glucose, Critically ill patients, Insulin-resistance, Insulin-glucose dynamics, Insulin sensitivity, Model-based control, Tight glycemic control €€€€,
"Abstract[swe]": €€€€
Tight glycemic control för kritiskt sjuka patienter på intensivvårdsavdelning är avgörande för att förebygga ogynnsamma utfall som påverkas av metabola fluktuationer i samband med insulinresistens. Traditionell intensiv insulinbehandling har visats inte tillräckligt kunna upprätthålla glukoskoncentrationen inom ett säkert och stabilt intervall. Genom att använda kliniskt validerade modeller möjliggörs skattning av okända insulin–glukos dynamiker med hjälp av tillståndsskattningsmetoder. Tidigare arbete med skattning av insulin–glukos-dynamik tog inte hänsyn till en divers syntetisk patientkohort med olika nutritionsinsatser, och motiveringen för parameteriseringen av det föreslagna filtret var bristfälligt

beskriven. I detta projekt utvecklades ett ramverk för att sampla syntetiska patienter som modellerar kritiskt sjuka patienter vid Karolinska Universitetssjukhuset, vilket möjliggjorde skattning av insulin–glukos dynamik. Intensive Control Insulin-Nutrition-Glucose modellen som användes i denna studie och i tidigare arbete visade sig inte vara observerbar vid beräkning av de empiriska observerbarhetsgramianerna, vilket ytterligare belyser begränsningarna med att modellera glukosreglering enbart utifrån blodglukosmätningar. Det identifierades att det utökade Kalmanfiltret hade svårigheter att skatta blodglukos när den initiala prediktionen avvek kraftigt från det sanna tillståndet ($\pm 10\%$ och $\pm 25\%$) jämfört med partikelfiltret, men att det ändå kunde köras betydligt snabbare. Ett partikelfilter med 750 partiklar uppnådde den bästa avvägningen mellan noggrannhet och komplexitet och gav mer än en sexfaldig minskning i komplexitet jämfört med den tidigare metoden med 5000 partiklar. För att vidareutveckla tidigare arbete skattades även en parameter som inte ingår i Intensive Control Insulin-Nutrition-Glucose modellens formulering. Partikelfiltret lyckades uppskatta både konstant och tidsvarierande insulinkänslighet, medan den utökade Kalmanfiltret endast tillförlitligt uppskattade den förra. Detta belyser begränsningarna hos de linjäriseringsapproximationer som används i det utökade Kalmanfiltret. Metodiken som introduceras i denna studie kan hjälpa kliniker att följa en systematisk process för att utveckla egna modellbaserade metoder för tight glycemic control.
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